DP Biology – B2.1

Membranes and membrane transport

- How do lipid and protein molecules come together to form membranes?
- What decides whether a substance can cross a membrane?

SL statements B2.1.1–10. Textbook pp. 223–237.

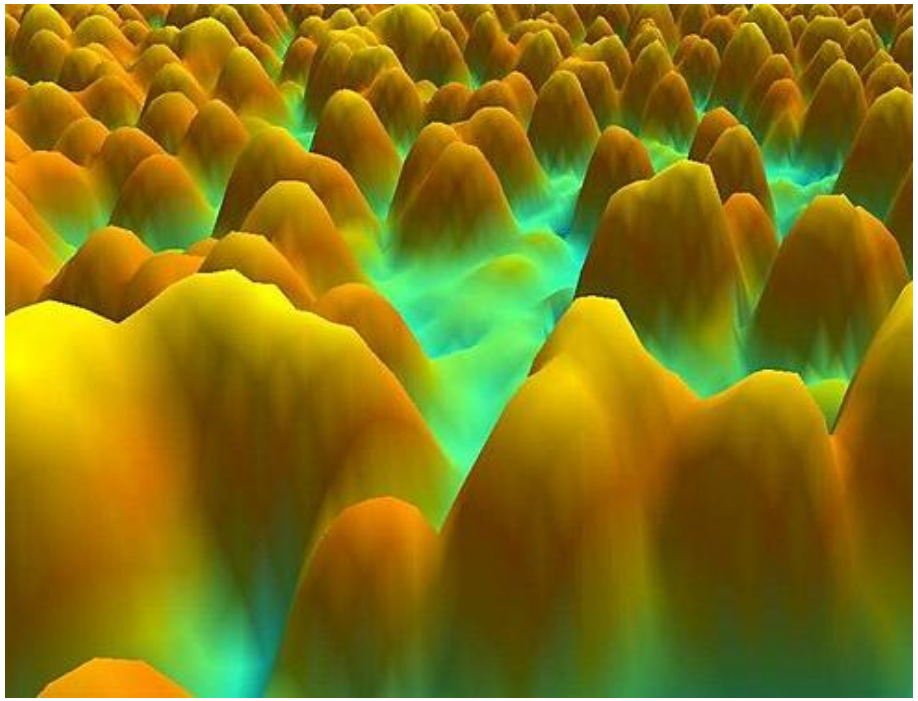


Image: a cell membrane with proteins, seen by atomic force microscopy. CC0.

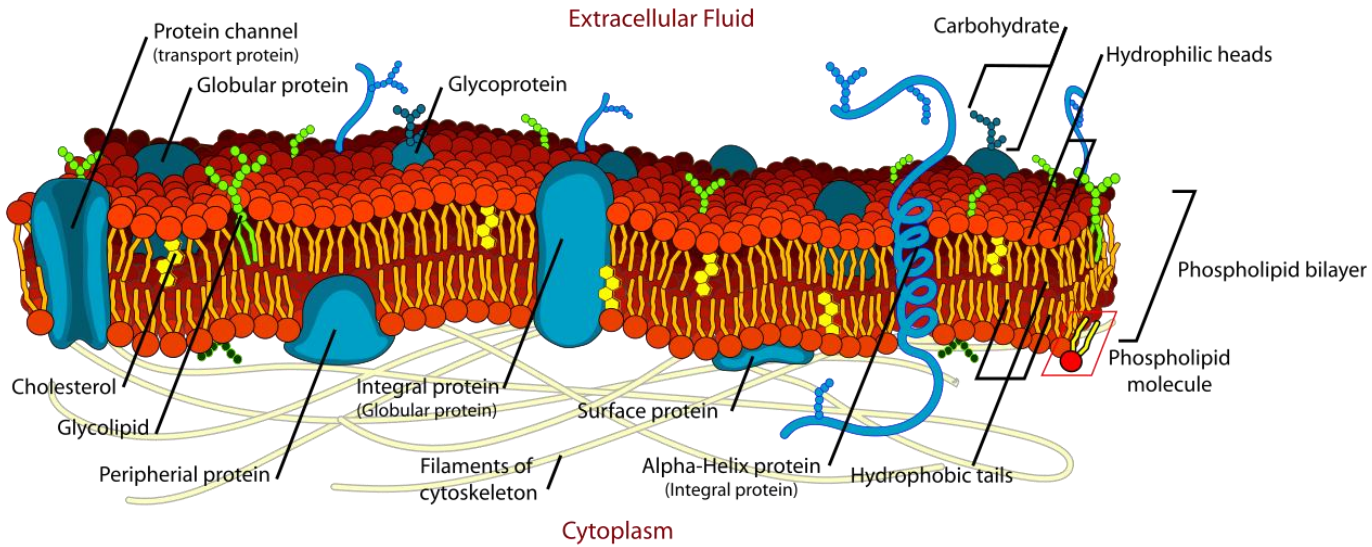
Objectives for each class

Class	Statements	By the end of the class, you can…
1	B2.1.1–4	Explain how phospholipids form a bilayer, why it is a barrier, simple diffusion, and membrane proteins
2	B2.1.5–7	Explain osmosis and aquaporins, facilitated diffusion through channels, and active transport by pumps
3	B2.1.8–10	Explain selective permeability, glycoproteins and glycolipids, and the fluid mosaic model

A wall only two molecules thick



Membrane proteins (AFM). CC0.



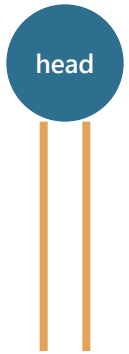
Cell membrane. LadyofHats, public domain.

The **plasma membrane** is a double layer of lipids with proteins floating in it. It holds the cell together, yet lets the right substances in and out.

Membranes also surround organelles, such as the nucleus and mitochondria.

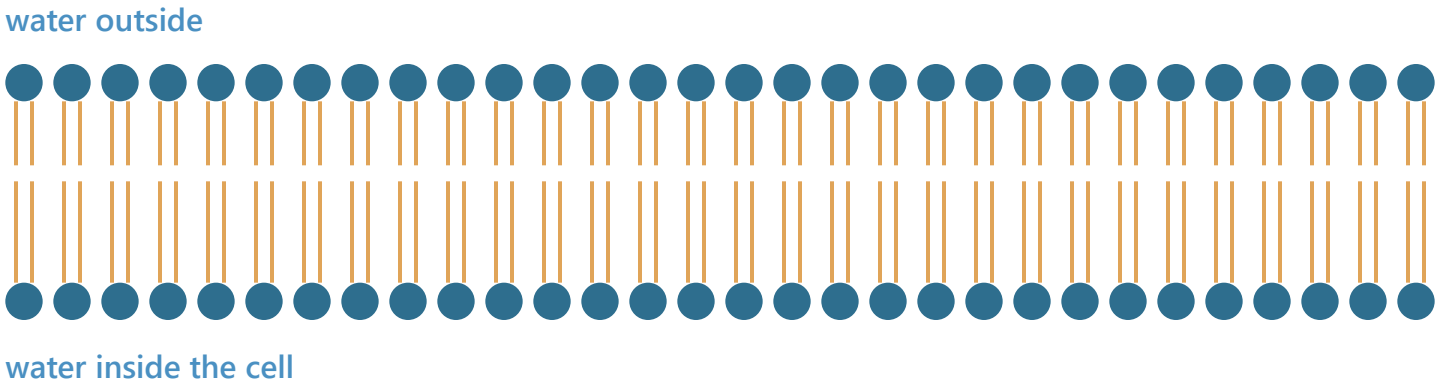
Phospholipids form a bilayer

B2.1.1



phosphate head: hydrophilic

fatty acid tails: hydrophobic

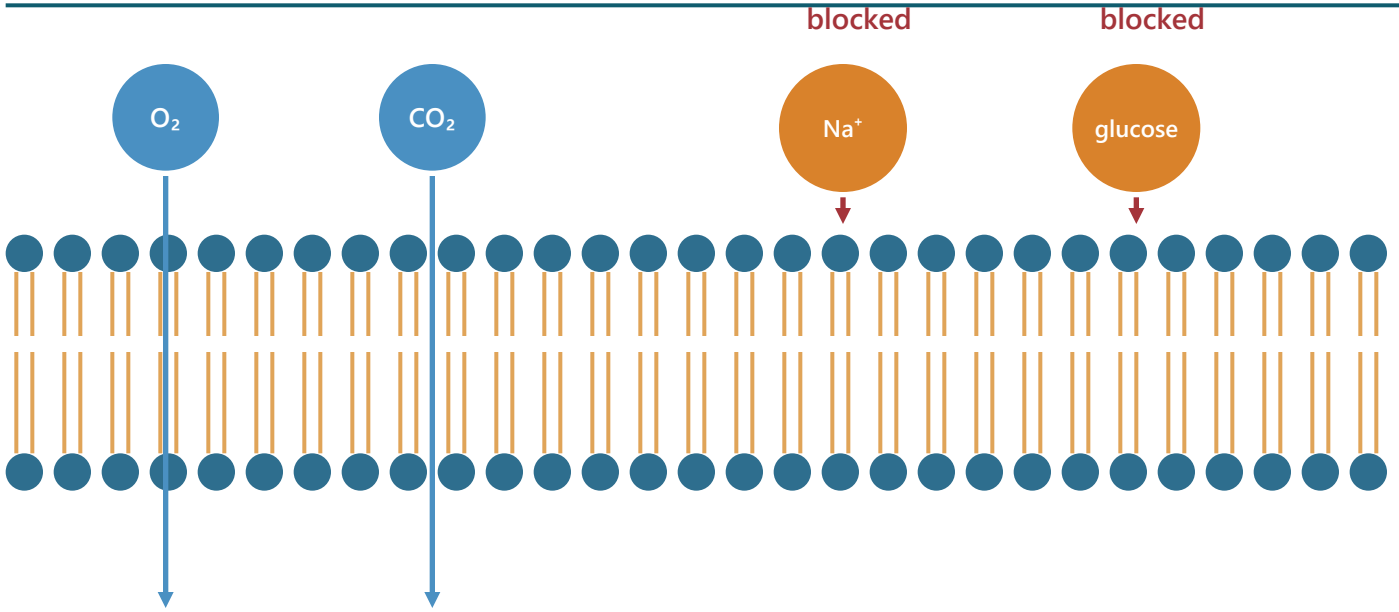


A phospholipid is **amphipathic**: its head is attracted to water but its tails repel water.

In water, phospholipids arrange themselves so the heads face the water on both sides and the tails hide in the middle. This **bilayer** forms by itself; it needs no energy.

Why the bilayer is a barrier

B2.1.2

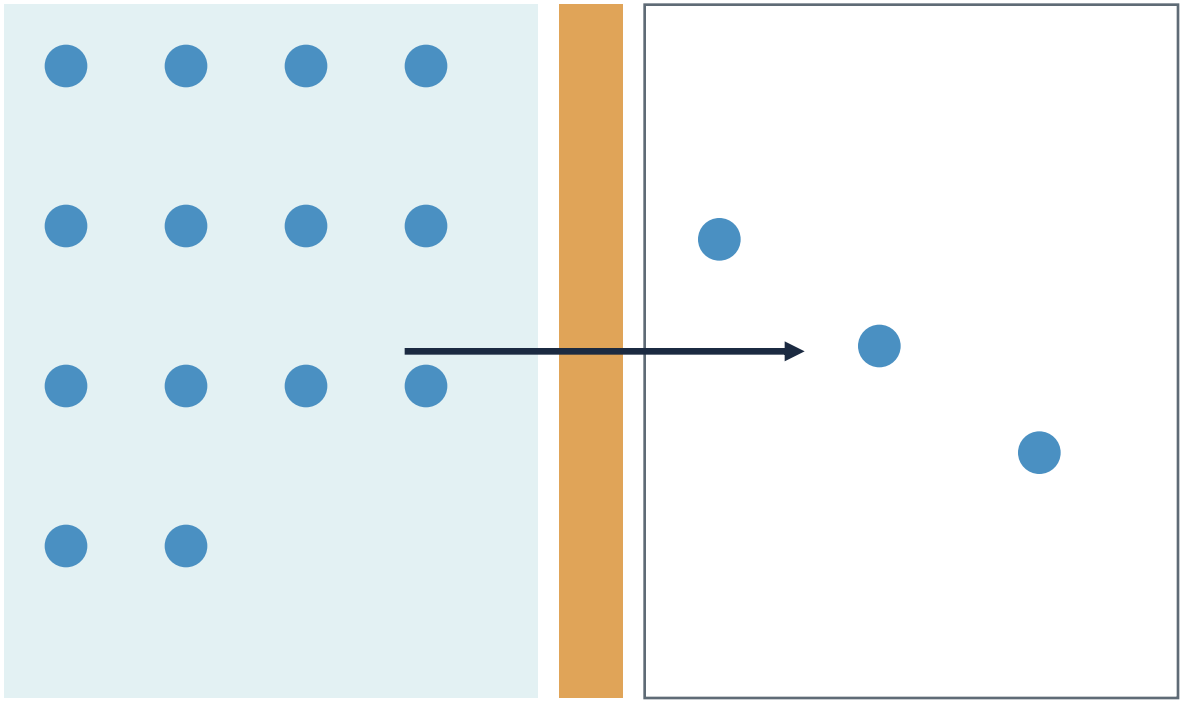


The middle of the bilayer is **hydrophobic**.

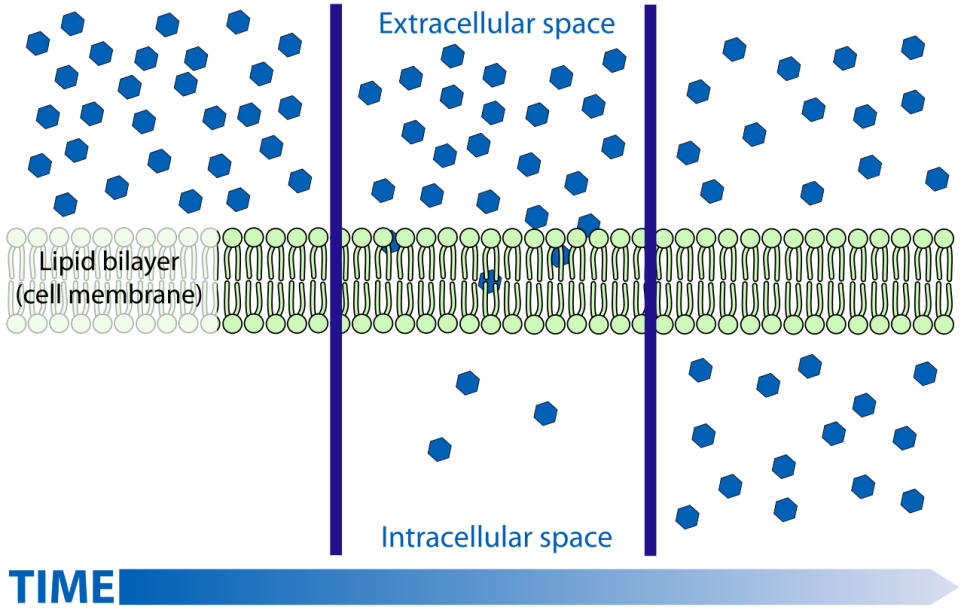
Small, non-polar molecules such as oxygen and carbon dioxide slip through easily.

Ions and large **polar** molecules such as glucose cannot dissolve in the tails, so they are blocked.

Simple diffusion



high concentration → low concentration



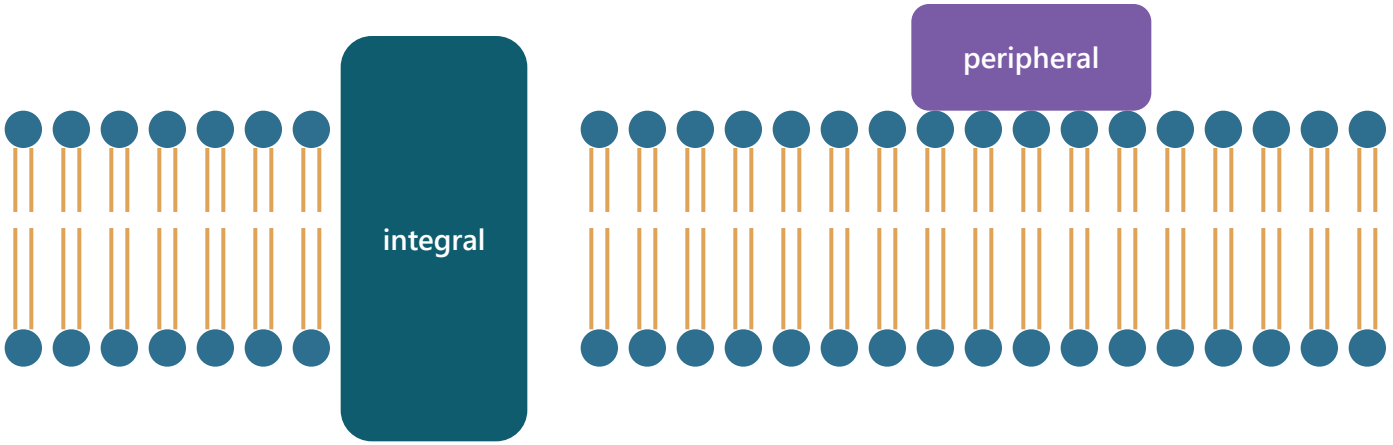
Simple diffusion. Public domain.

Particles spread out from high to low concentration by random movement. No energy and no protein are needed.

Steeper gradient = faster diffusion.

Integral and peripheral proteins

B2.1.4



Integral proteins pass into or through the bilayer; peripheral ones sit on the surface.

Job	Example
Channel	ion channels
Pump	Na^+/K^+ pump
Receptor	hormone receptor
Enzyme	in the mitochondrion
Recognition	glycoproteins

Why integral proteins stay put

Their middle section has hydrophobic amino acids that sit among the tails; their ends are hydrophilic, facing the water.

Practice: which can cross?

B2.1.2–3

Substance	Size	Polar or charged?
Oxygen	small	no
Ethanol	small	slightly
Glucose	large	polar
Sodium ion	small	charged
Steroid hormone	large	no

1. Predict which can diffuse directly through the bilayer.
2. Explain your choice for sodium ions.
3. Suggest how glucose enters cells.

Check-in: class 1

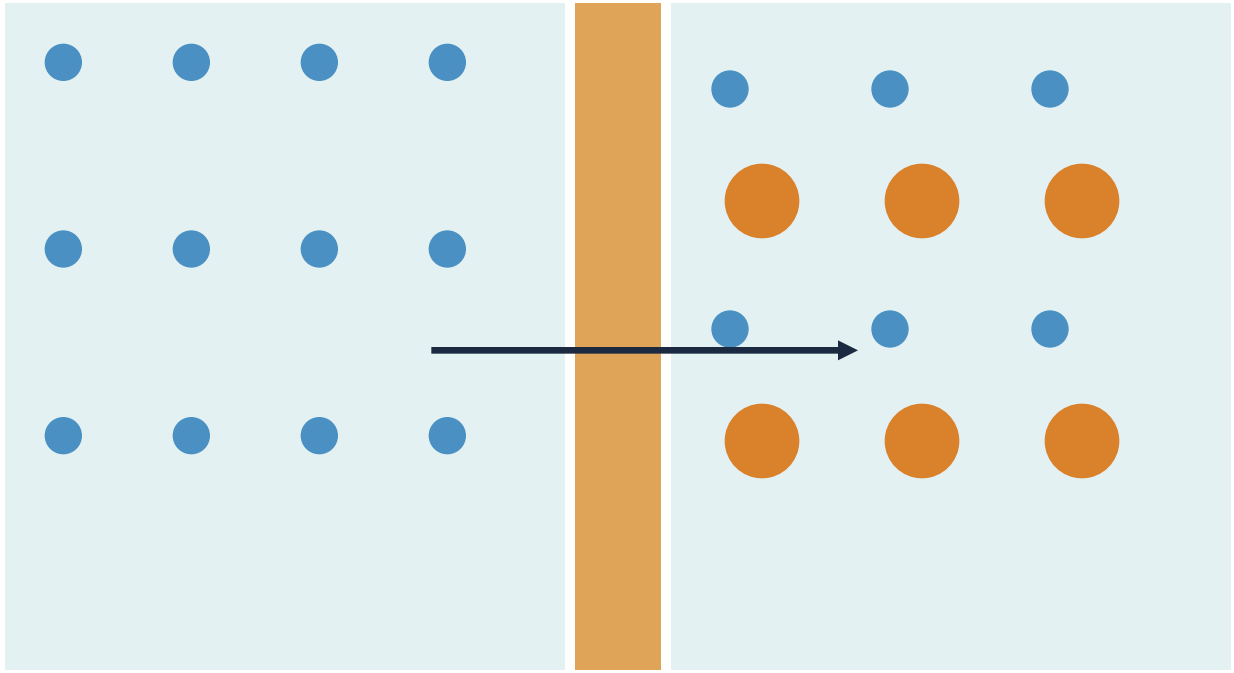
B2.1.1–4

1. Explain why phospholipids form a bilayer in water.
2. Explain why ions cannot pass through the bilayer.
3. Define simple diffusion.
4. Distinguish between integral and peripheral proteins.
5. Give three jobs of membrane proteins.

How to use this

Answer without your notes, then check against slides 4–8.

Osmosis: water on the move



more water, less solute

less water, more solute

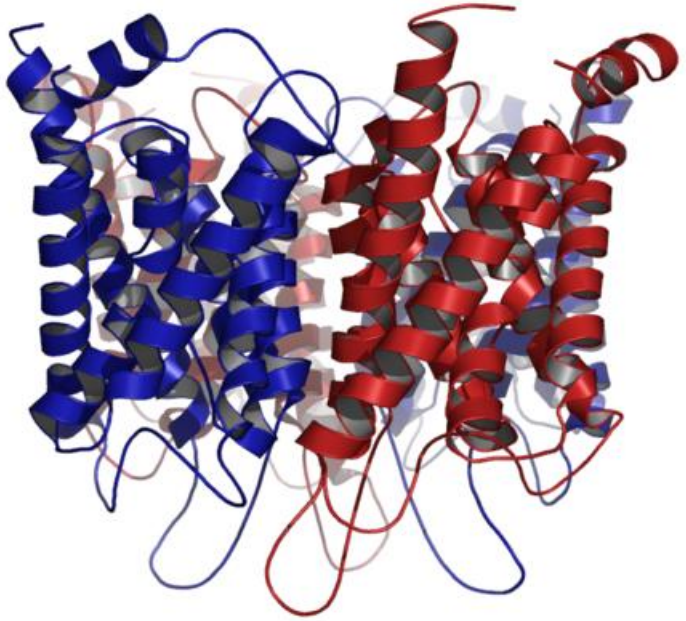
Osmosis is the net movement of **water** across a membrane, from a lower to a higher **solute** concentration.

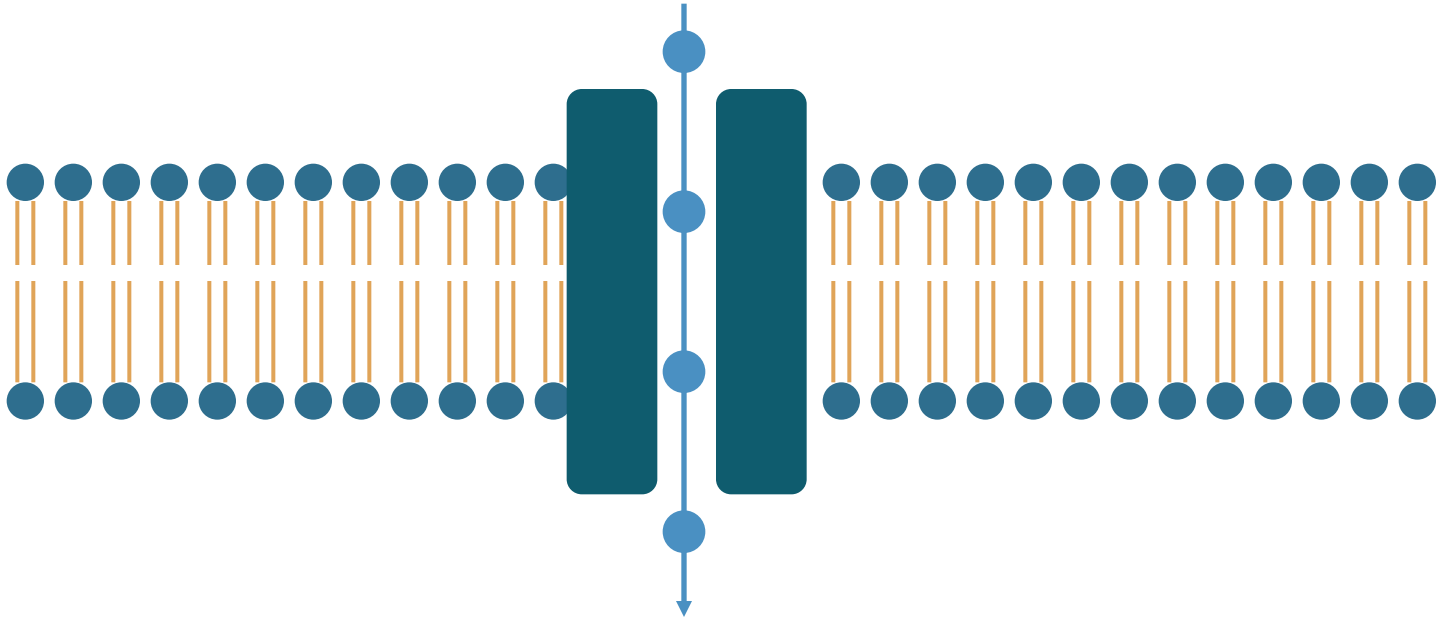
The solute particles cannot cross, but water can, so water moves towards the side with more solute.

Remember

Osmosis is about water only. The solute is what draws the water across.

Aquaporins: water channels





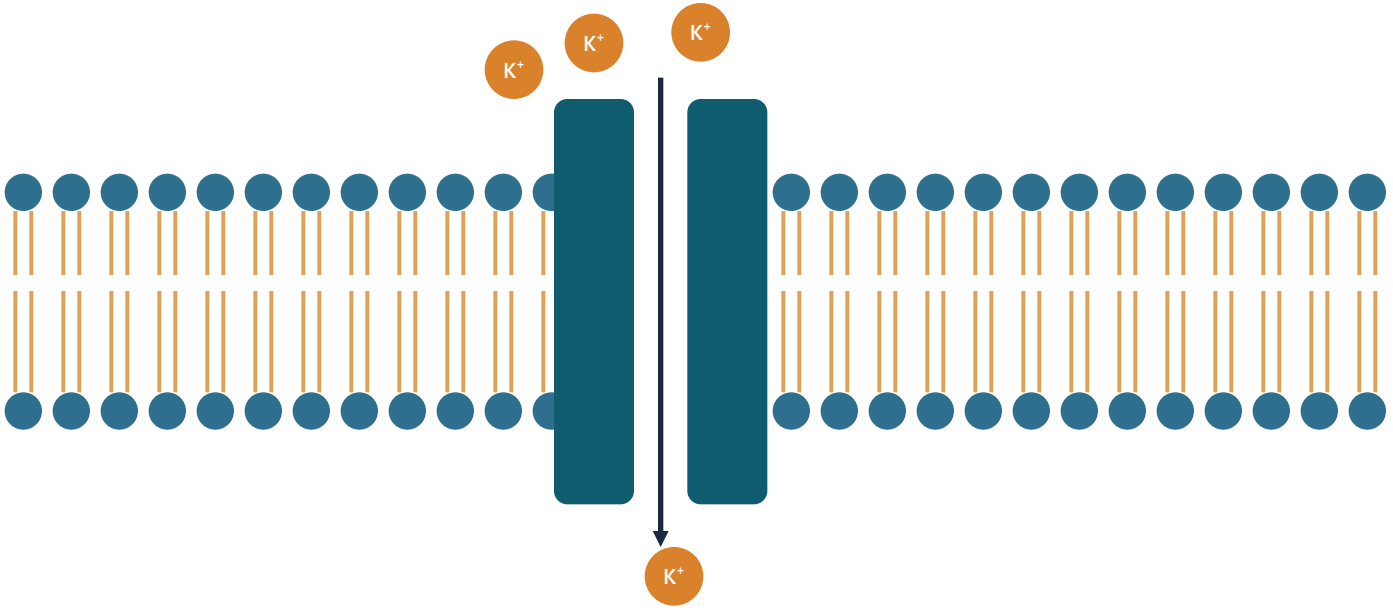
Aquaporins are channel proteins that let water through in single file, many times faster than through the bilayer.

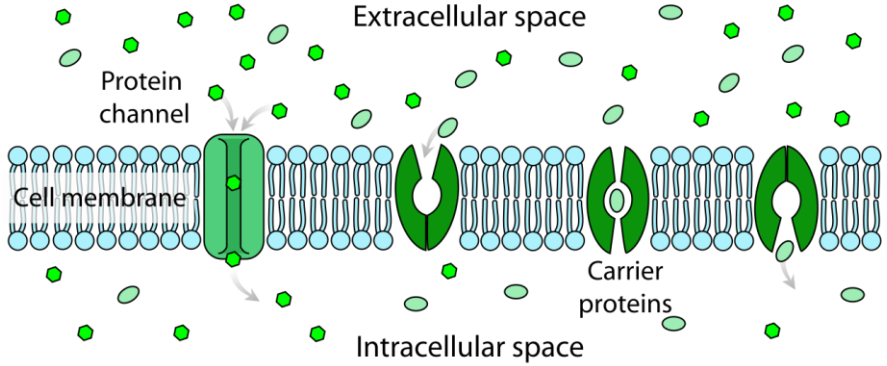
Kidney cells add aquaporins to their membranes to reabsorb water.

Aquaporin. CC BY-SA 3.0.

Facilitated diffusion through channels

B2.1.6





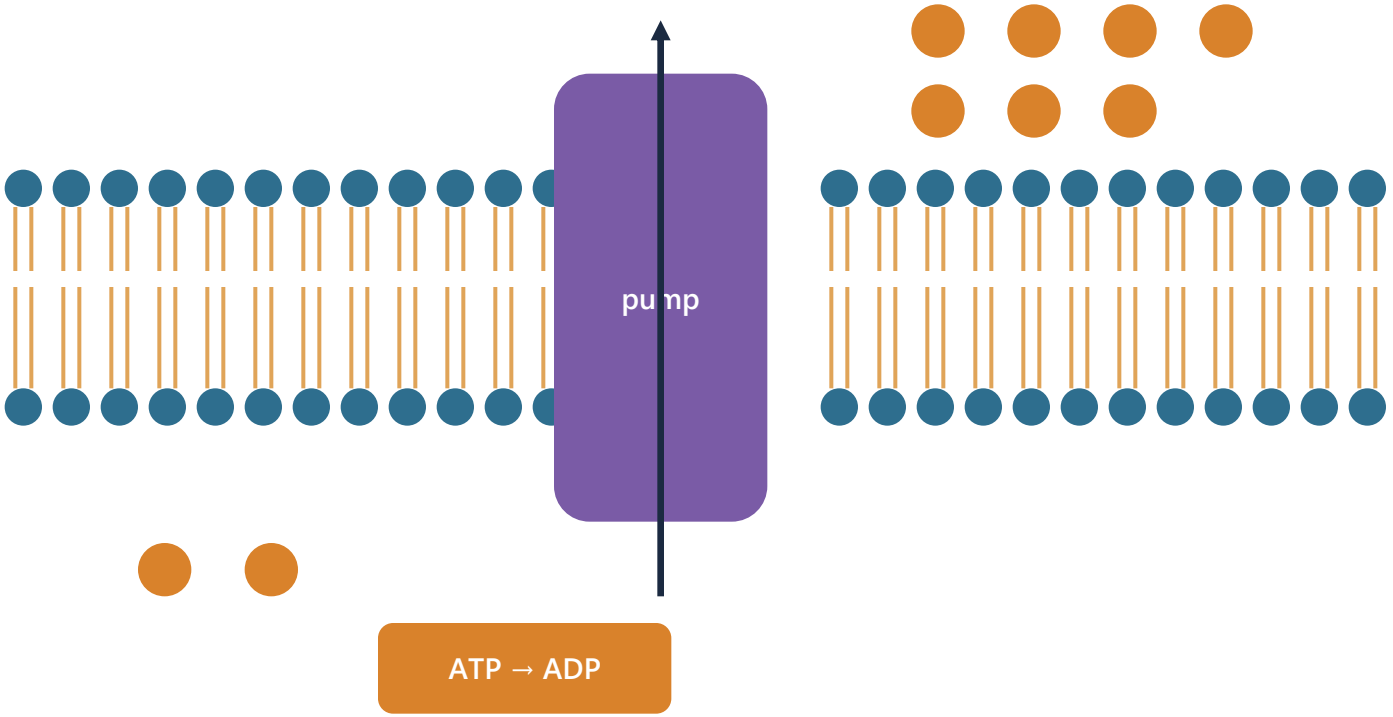
Facilitated diffusion. Public domain.

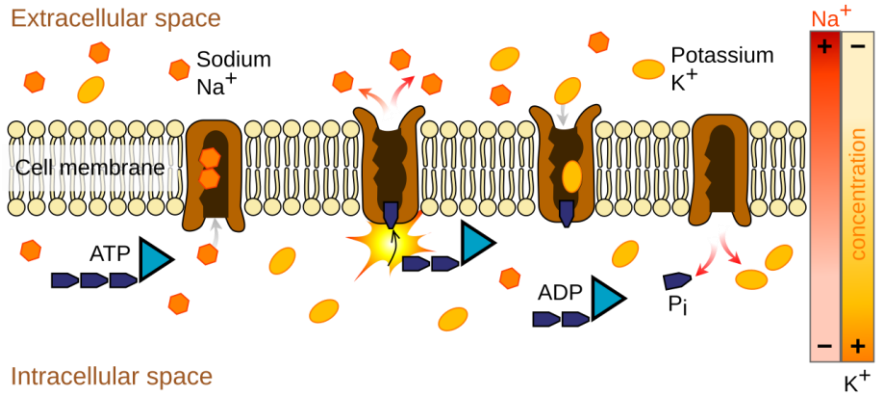
Channel proteins make a hydrophilic pore through the membrane. Each channel lets only one kind of particle through, decided by its size and charge.

It is still **diffusion**: down the concentration gradient, no energy used.

Pumps: active transport

B2.1.7





Sodium-potassium pump. Public domain.

Pump proteins use energy from **ATP** to move particles **against** their concentration gradient, from low to high.

This lets cells build up substances they need, such as mineral ions in root hair cells.

Four ways across

B2.1.3, 5–7

	Through what	Gradient	Energy
Simple diffusion	the bilayer	down	none
Osmosis (water)	bilayer and aquaporins	to higher solute	none
Facilitated diffusion	channel proteins	down	none
Active transport	pump proteins	against	ATP

Key idea

Only active transport can move particles against their gradient, so only it needs ATP.

Working with data: glucose uptake

B2.1.6

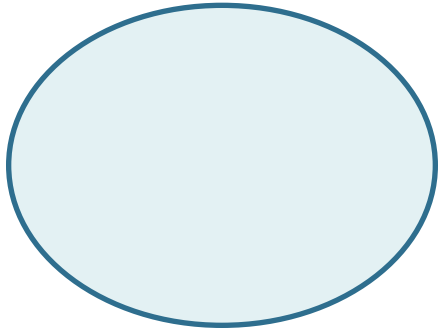
Rate of glucose uptake by red blood cells at different outside glucose concentrations. These data are invented for practice; the pattern is realistic.

Glucose / mmol dm^{-3}	Uptake rate
0	0
2	40
5	72
10	92
20	98

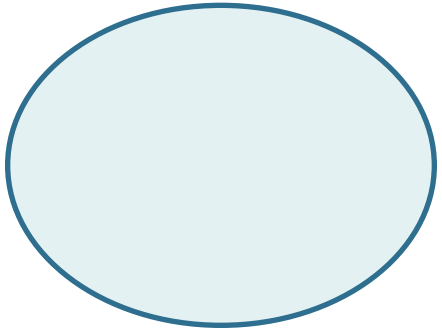
1. Describe the pattern.
2. Explain why the rate levels off.
3. Predict the shape of the graph if glucose crossed by simple diffusion.

Practice: which way will water move?

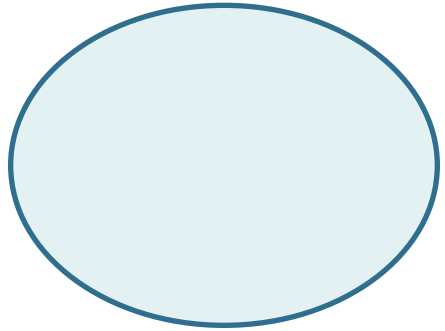
B2.1.5



Cell 2% salt, outside 5% salt



Cell 3% sugar, outside pure water



Cell 1% salt, outside 1% salt

1. For each cell, state whether water moves in, moves out, or shows no net movement.
2. Explain one of your answers using the term solute concentration.

Check-in: class 2

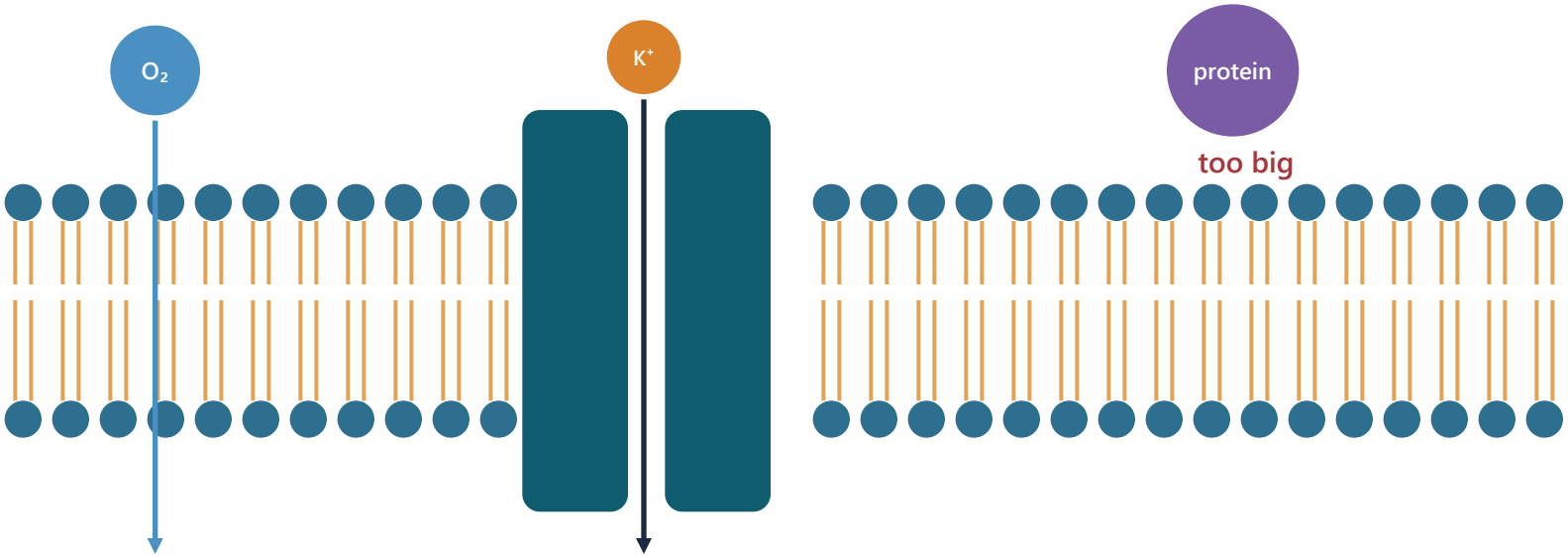
B2.1.5–7

1. Define osmosis.
2. Explain how aquaporins increase the rate of osmosis.
3. Distinguish between facilitated diffusion and active transport.
4. Explain why channel proteins are selective.
5. Explain why only active transport needs ATP.

How to use this

Answer without your notes, then check against slides 10–16.

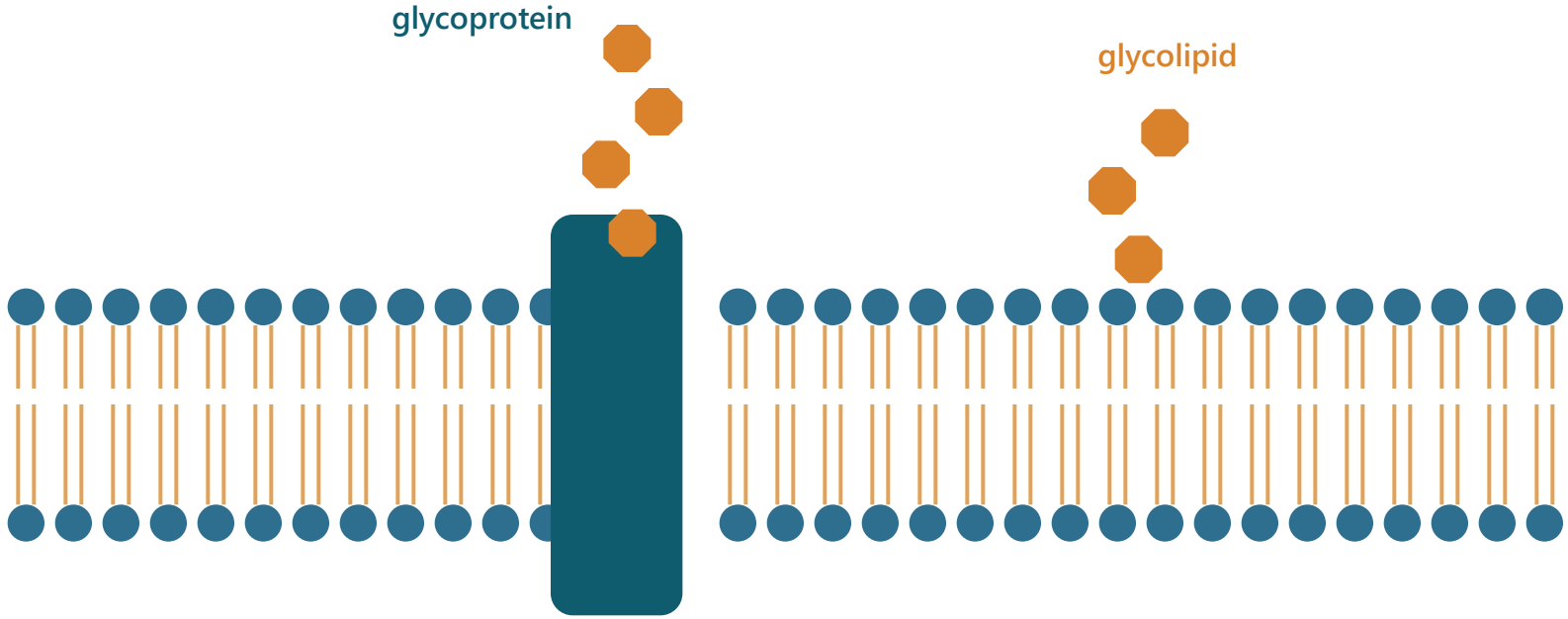
Selectively permeable



Membranes let some substances through easily, others only through specific proteins, and some not at all.

So the cell controls what enters and leaves: it is **selectively permeable**.

Glycoproteins and glycolipids

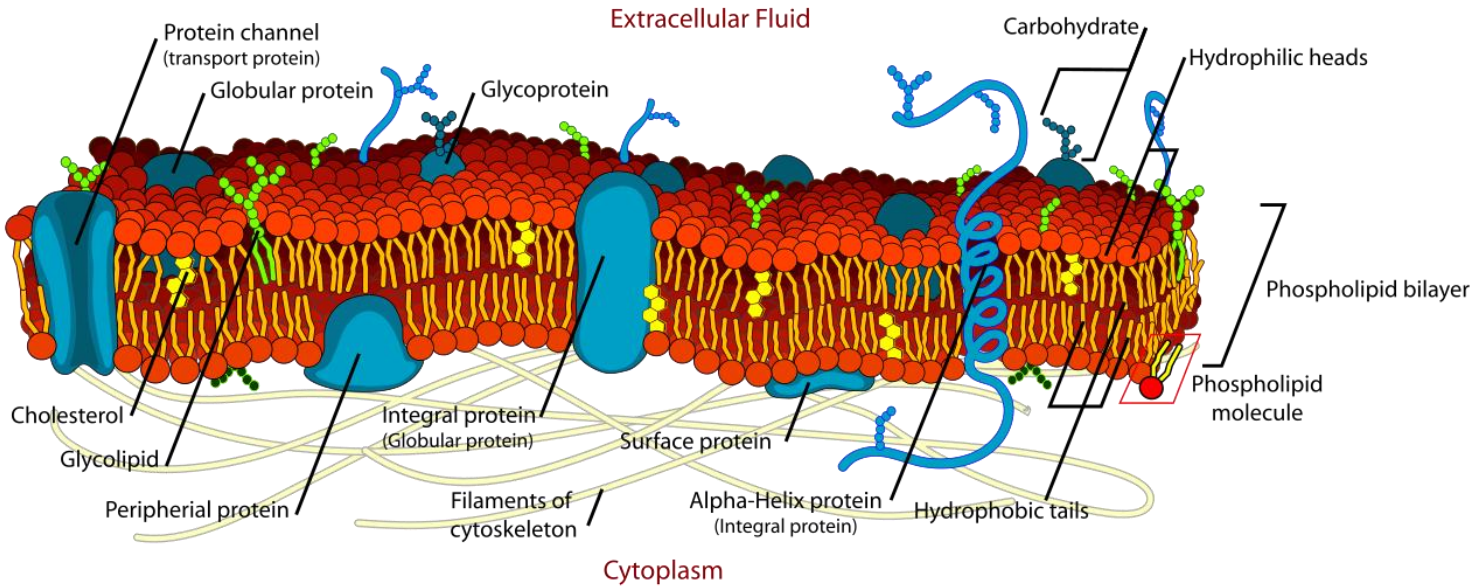


Short carbohydrate chains are attached to proteins or lipids on the **outside** of the membrane.

Together they form the **glycocalyx**: used for cell recognition (for example blood groups and immune cells) and for cells sticking together.

The fluid mosaic model

B2.1.10



Fluid: phospholipids and many proteins move sideways freely.

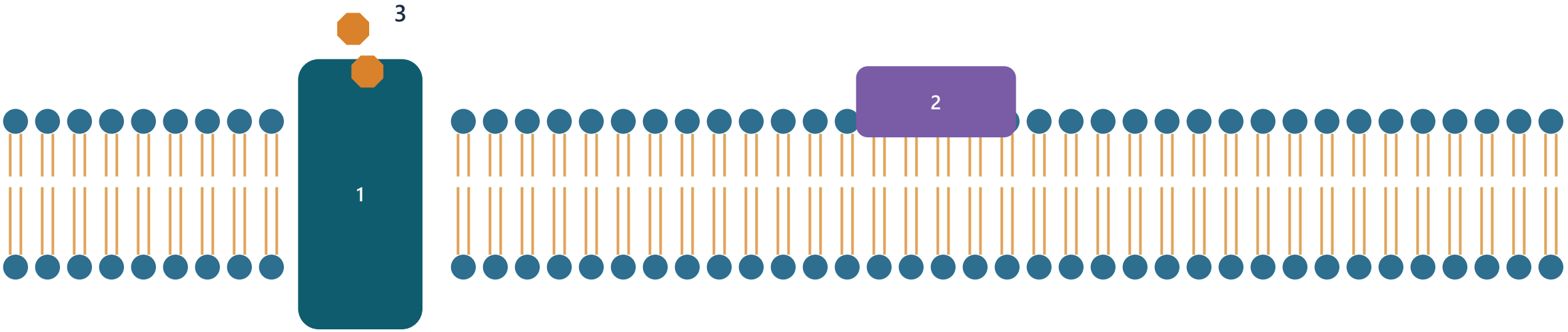
Mosaic: proteins are scattered through the bilayer like tiles.

Be ready to draw it, labelling the bilayer, heads, tails, integral and peripheral proteins, glycoproteins, glycolipids and cholesterol.

Cell membrane. LadyofHats, public domain.

Practice: draw the fluid mosaic model

B2.1.10



1. Name structures 1, 2 and 3.
2. Label a hydrophilic head and a hydrophobic tail.
3. Add a glycolipid and a channel protein to your own drawing.
4. Explain why the model is called fluid mosaic.

Common mistakes to avoid

Not this	But this
Osmosis is the movement of solute.	Osmosis is the movement of water.
Facilitated diffusion uses energy.	It is passive; only pumps use ATP.
Active transport moves particles down the gradient.	It moves them against the gradient.
Proteins are fixed in the membrane.	Most can move sideways: the membrane is fluid.

Check that you can...

1. **Explain** how phospholipids form a bilayer, and why it is a barrier
2. **Define** simple diffusion and osmosis
3. **Distinguish** integral and peripheral proteins
4. **Explain** how aquaporins and channels work
5. **Compare** passive and active transport
6. **Outline** the roles of glycoproteins and glycolipids
7. **Draw** and **label** the fluid mosaic model

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why phospholipids form a bilayer in water.
2. Distinguish between facilitated diffusion and active transport.
3. State two components of the fluid mosaic model and their roles.

Image credits

Cell membrane with proteins visualized by AFM.jpg – Nucturra – CC0

Cell membrane detailed diagram en.svg – Mariana Ruiz – Public domain

Scheme simple diffusion in cell membrane-en.svg – LadyofHats Mariana Ruiz Villarreal – Public domain

Aquaporin-Sideview.png – Vossman – CC BY-SA 3.0

Scheme facilitated diffusion in cell membrane-en.svg – LadyofHats Mariana Ruiz Villarreal – Public domain

Scheme sodium-potassium pump-en.svg – LadyofHats Mariana Ruiz Villarreal – Public domain

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